

POWELL RIVER, BC, Canada

WMO: 717200

Lat: 49.8344N

Lon: 124.4981W

Elev: 125

StdP: 99.83

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.1	-3.5	-10.7	1.5	-2.9	-7.1	2.1	-1.4	11.3	6.8	10.1	6.2	0.9	90	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	27.0	17.9	25.1	17.2	23.4	16.6	18.7	25.0	17.9	23.6	17.1	22.0	2.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.8	20.8	15.6	11.3	20.0	14.9	10.8	19.3	53.4	25.3	50.8	23.7	48.4	22.1	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.6	6.5	DB	-8.8	30.4	2.0	2.2	-10.2	32.0	-11.3	33.3	-12.4	34.5	-13.9	36.1	(4)
(5)			WB	-9.4	20.4	1.9	1.1	-10.8	21.2	-11.9	21.8	-13.0	22.4	-14.4	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	3.4	3.2	5.3	7.8	12.2	14.8	17.6	17.5	14.0	9.1	5.4	2.7	(6)
(7)		DBStd	6.00	3.11	3.07	2.47	2.35	2.72	2.56	2.23	2.16	2.59	2.42	3.22	3.05	(7)
(8)		HDD10.0	1041	206	190	147	74	10	1	0	0	1	46	141	227	(8)
(9)		HDD18.3	3283	464	423	404	317	189	111	40	43	132	286	388	485	(9)
(10)		CDD10.0	838	0	0	1	7	79	144	235	232	121	18	3	0	(10)
(11)		CDD18.3	39	0	0	0	0	0	5	17	16	2	0	0	0	(11)
(12)		CDH23.3	389	0	0	0	0	11	58	158	139	23	0	0	0	(12)
(13)		CDH26.7	67	0	0	0	0	0	11	32	23	2	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.4	2.2	2.8	2.7	2.4	2.5	2.5	2.2	2.2	2.5	2.8	2.5	(14)
(15)	Precipitation	PrecAvg	1136	144	105	100	64	66	54	45	51	62	131	162	160	(15)
(16)		PrecMax	1436	296	218	186	122	123	120	103	182	159	268	273	230	(16)
(17)		PrecMin	685	33	44	45	12	24	10	0	2	4	27	61	44	(17)
(18)		PrecStd	170	66	43	37	27	28	29	25	41	37	62	62	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	10.7	15.2	19.3	25.0	28.2	30.0	29.2	26.2	17.5	13.5	10.8	(19)
(20)			MCWB	9.6	9.0	9.6	12.5	14.7	17.3	19.3	18.5	17.4	13.6	11.7	9.7	(20)
(21)		2%	DB	9.5	9.6	12.0	15.6	22.2	24.9	26.9	26.7	22.9	15.5	12.1	8.9	(21)
(22)			MCWB	8.3	8.0	8.7	10.0	14.6	16.4	18.0	18.1	16.2	12.9	10.7	7.7	(22)
(23)		5%	DB	8.6	8.6	10.5	13.7	20.0	22.1	24.9	24.7	20.6	14.2	11.0	7.9	(23)
(24)			MCWB	7.6	7.2	7.9	9.1	14.0	15.4	17.3	17.7	15.5	12.1	9.9	7.0	(24)
(25)		10%	DB	7.7	7.5	9.3	12.2	18.0	19.9	23.1	22.8	18.8	13.1	9.9	7.1	(25)
(26)			MCWB	6.7	6.3	7.3	8.4	12.9	14.3	16.7	16.8	14.6	11.3	8.7	6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.9	10.8	13.1	16.2	18.7	20.3	19.7	18.6	14.8	12.5	10.0	(27)
(28)			MCDWB	10.5	10.4	12.8	17.4	22.5	25.6	29.1	26.4	24.3	15.8	13.4	10.6	(28)
(29)		2%	WB	8.6	8.6	9.3	11.3	15.0	17.1	18.8	18.7	17.1	13.7	10.9	8.1	(29)
(30)			MCDWB	9.1	9.2	11.3	14.3	20.5	23.7	25.1	24.9	21.2	15.0	11.7	8.7	(30)
(31)		5%	WB	7.7	7.3	8.3	10.0	14.2	15.8	17.8	17.9	16.1	12.6	10.0	7.1	(31)
(32)			MCDWB	8.5	8.1	10.1	12.4	19.0	20.7	23.4	23.6	19.6	13.8	10.9	7.8	(32)
(33)		10%	WB	6.8	6.5	7.5	9.1	13.4	14.9	17.0	17.2	15.1	11.4	8.9	6.2	(33)
(34)			MCDWB	7.5	7.4	9.1	11.2	17.3	19.0	22.1	21.8	17.9	12.7	9.8	7.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	5.6	6.8	8.1	9.5	8.8	10.1	10.1	8.5	6.3	5.2	4.2	(35)
(36)			MCDBR	4.7	6.2	8.6	11.4	14.0	13.4	13.8	13.8	12.4	8.0	4.7	3.8	(36)
(37)			MCWBR	3.9	4.5	5.5	6.6	7.2	6.1	6.2	6.3	6.3	5.3	3.8	3.2	(37)
(38)		5% WB	MCDWR	4.4	4.8	7.6	8.7	12.1	11.5	12.7	12.5	10.5	6.3	4.3	3.8	(38)
(39)			MCWBR	3.8	3.7	5.2	5.4	6.4	5.6	5.9	6.1	5.6	4.5	3.6	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.308	0.333	0.353	0.374	0.373	0.369	0.369	0.340	0.337	0.317	0.307	(40)	
(41)		taud	2.421	2.510	2.447	2.414	2.397	2.424	2.470	2.473	2.557	2.524	2.491	2.410	(41)	
(42)		Ebn at Noon	731	846	877	893	885	886	884	867	863	798	727	681	(42)	
(43)		Edh at Noon	63	73	92	106	113	111	104	99	82	72	59	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.80	1.67	2.48	3.89	5.09	5.48	6.04	5.19	3.61	1.93	0.94	0.62	(44)	
(45)		RadStd	0.12	0.28	0.34	0.41	0.56	0.62	0.65	0.44	0.32	0.33	0.18	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page